Solutions to Selected Problems on HW#3

Note: In pretty much any mathematical problem beyond the level of high school algebra, there will be many possible solutions and many ways to elegantly write up the solution. So while your solution need not look exactly like what's here, it should be direct and clearly written.

20 (pg. 56):

After some inspection, one sees that some of the listed subgroups are not comparable. So we in fact need two diagrams to describe all the containments:

$$\begin{tikzcd}[column sep=small, row sep=small]
 & G_4 \arrow[d, phantom, "\rotatebox{90}{$\leq$}"] & \\
 & G_1 \arrow[dl, phantom, "\rotatebox{45}{$\leq$}"] \arrow[dr, phantom, "\rotatebox{-45}{$\geq$}"] & \\
G_7 \arrow[dr, phantom, "\rotatebox{-45}{$\geq$}"] & & G_9 \arrow[dl, phantom, "\rotatebox{45}{$\leq$}"] \\
 & G_2 &
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}[column sep=small, row sep=small]
 & G_5 \arrow[dl, phantom, "\rotatebox{45}{$\leq$}"] \arrow[dr, phantom, "\rotatebox{-45}{$\geq$}"] & \\
G_3 \arrow[d, phantom, "\rotatebox{90}{$\leq$}"] & & G_6 \\
G_9 & &
\end{tikzcd}$$

The containments on the left are based on the simple fact that $a|b \Longrightarrow \langle b\rangle \subseteq \langle a\rangle$, within $\langle \mathbb{Z}, +\rangle$. (So the bigger the number, the smaller the subgroup.) The containments on the right are based on four even simpler facts: (i) every rational number is a real number, (ii) exponents add under multiplication of powers, (iii) π is a real number, and (iv) 6 is a rational number. ■

18 (pg. 66):

Although one could simply count by listing all the multiples of 30 mod 42, we can save time via one of our theorems last week (Theorem 0.2 in the hand-out from Item #13 of `www.math.tamu.edu/~rojas/hqf415.html`): The second part of the theorem (with $g=1$, $n=42$,and $k=30$) tells us that $\#\langle 30\rangle = \frac{42}{\gcd(30,42)} = \frac{42}{6} = 7$. So the number of elements is 7. ■

16 (pg. 73):

The Cayley diagram for $\mathbb{Z}/8\mathbb{Z}$, with generating set $\{2, 5\}$ can be drawn in many (equally correct) ways. Here are two: